# Freight rate prediction assessment

Reproducible model-development and validation record

This report follows the order in which the solution was developed: establish the deployment boundary, audit the data, form testable hypotheses, run forward experiments, evaluate once on a locked holdout, and only then train on all labeled data. This order prevents a persuasive explanation from being invented after seeing a preferred result.

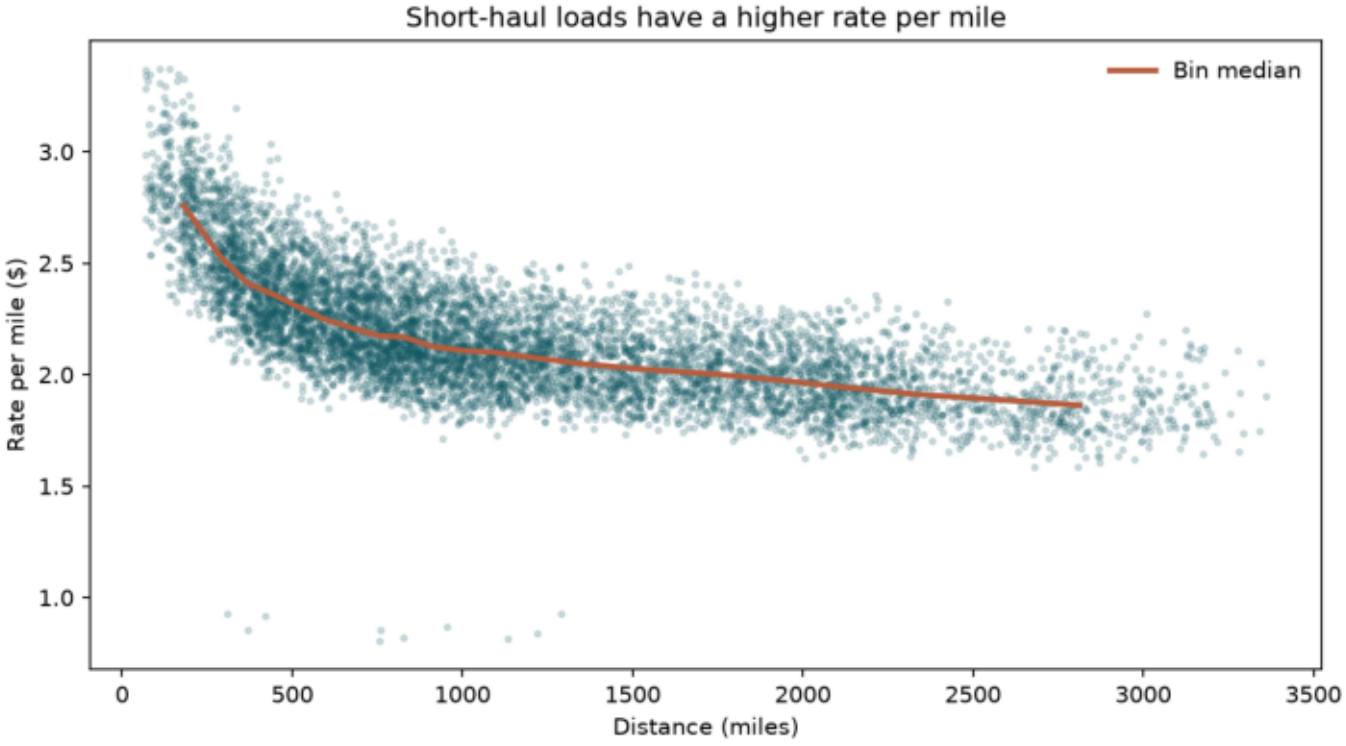
The final task is future-period regression. The labeled data ends on October 31, while every hidden evaluation row falls in November or December. Consequently, model selection used complete-date forward splits rather than random cross-validation.

# 1. Audit findings and resulting questions

The schema audit found 48,000 labeled rows and 12,000 final rows. Target-aware discovery used the 38,477 rows through August 31, preserving September and October as a locked holdout. In that discovery period, distance had Pearson correlation 0.910 with posted rate. There were 292 negative training weights, 300 missing weights, and 374 missing market indices. The final data introduced 8 cities absent from training, affecting 1,447 rows.

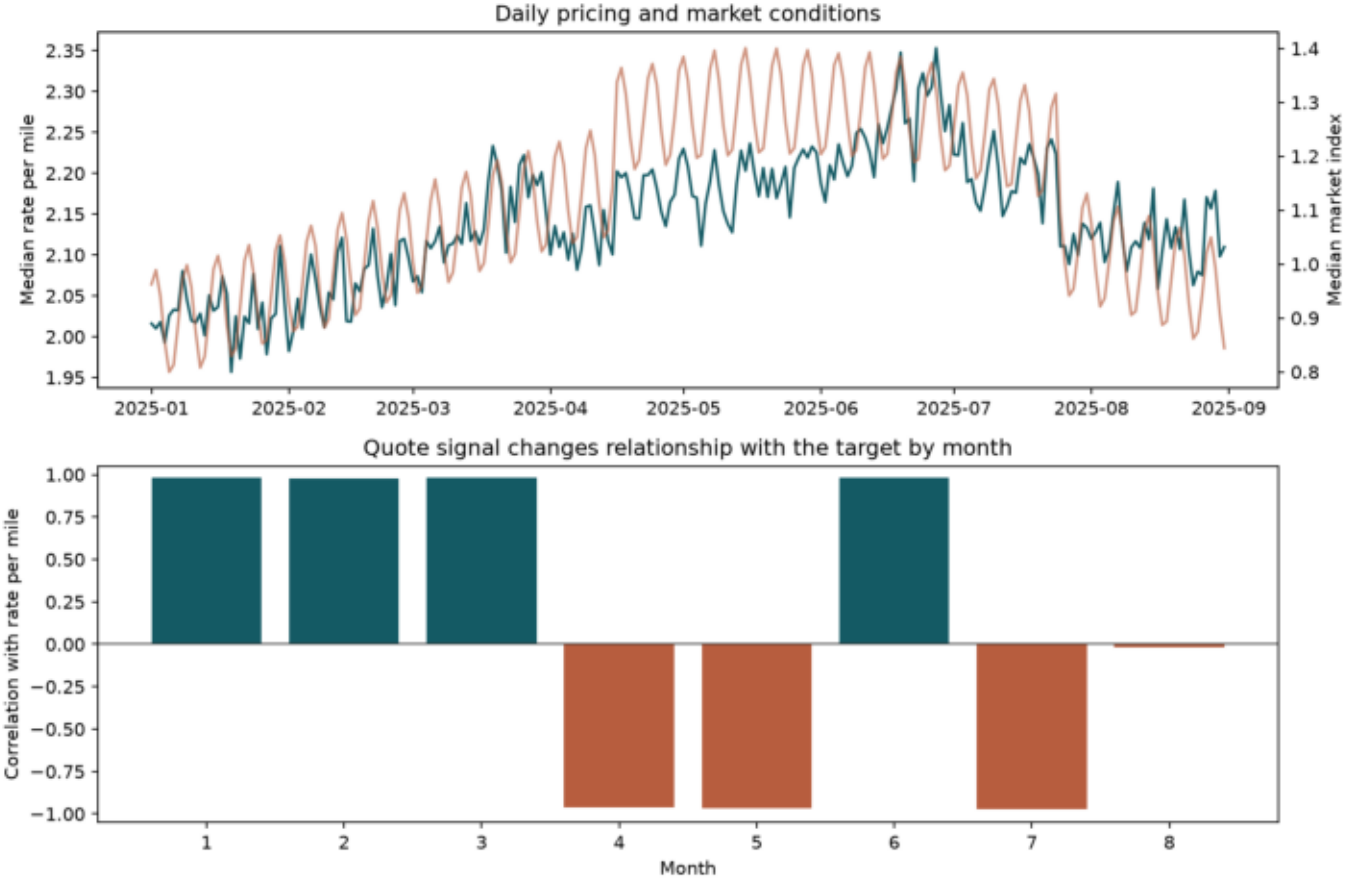
These observations generated four experiments rather than immediate conclusions. First, compare raw-rate and rate-per-mile targets. Second, test robust training because the target has isolated extreme RPM values. Third, compare coordinate-first and route-category models under held-out-city tests. Fourth, measure quote-signal stability month by month before deciding whether it is safe for future prediction.

# 2. Target structure




The median rate per mile falls with distance, consistent with fixed pickup and delivery costs being spread over more miles. This motivated rate per mile as a candidate target, but the choice was retained only if it beat direct-rate modeling on forward folds.

# 3. Temporal stability and leakage audit




The quote signal changes from strongly positive to strongly negative across months. Because final predictions occur in unseen future months, this sign reversal is a temporal stability failure. The quote feature was kept as an ablation candidate but excluded from the eligible final models.

# 4. Forward-fold model comparison

| candidate | mae | rmse | clean_mae | bias |
|---|---|---|---|---|
| Route CatBoost RPM | 117.79 | 650.43 | 63.91 | -33.39 |
| Structural CatBoost RPM | 120.67 | 650.53 | 66.84 | -30.25 |
| Structural CatBoost direct | 121.32 | 650.74 | 67.49 | -24.74 |
| Structural RPM + quote | 147.63 | 654.78 | 94.18 | -34.72 |
| Freight-rule baseline | 164.71 | 665.77 | 111.11 | -88.18 |
| Ridge RPM | 165.89 | 660.45 | 112.84 | 48.61 |
| Structural RPM squared loss | 171.95 | 684.52 | 118.87 | 42.12 |

Metrics are averages across June, July, and August expanding-window folds. MAE describes the ordinary dollar miss. RMSE is reported because the hidden metric is undisclosed, though it is dominated by apparently random target multipliers. The final candidate was selected using normalized MAE and RMSE against the same baseline, with worst-fold error as a tie-breaker.

# 5. Cold-start stress test

| candidate | mae | rmse | clean_mae |
|---|---|---|---|
| Route CatBoost RPM | 145.74 | 602.46 | 96.08 |
| Structural CatBoost RPM | 147.51 | 600.18 | 97.87 |

Three groups of eight cities were removed entirely from pre-July fitting data and evaluated on July and August rows. This leaves the September-October locked holdout untouched. Coordinates provide a prediction path for unseen cities. Category-heavy model weight was capped when it degraded held-out-city MAE by more than five percent.

# 6. Locked holdout and final model

After tuning, the selected primary candidate was Structural CatBoost RPM. The final blend assigned 40% to the selected primary model and 60% to the route-category model. No design choices were changed after the September-October holdout was scored.

Locked-holdout MAE: $119.36
Locked-holdout RMSE: $637.19
Locked-holdout median absolute error: $45.82
Locked-holdout bias: -$91.85
Locked-holdout clean-row MAE: $65.40

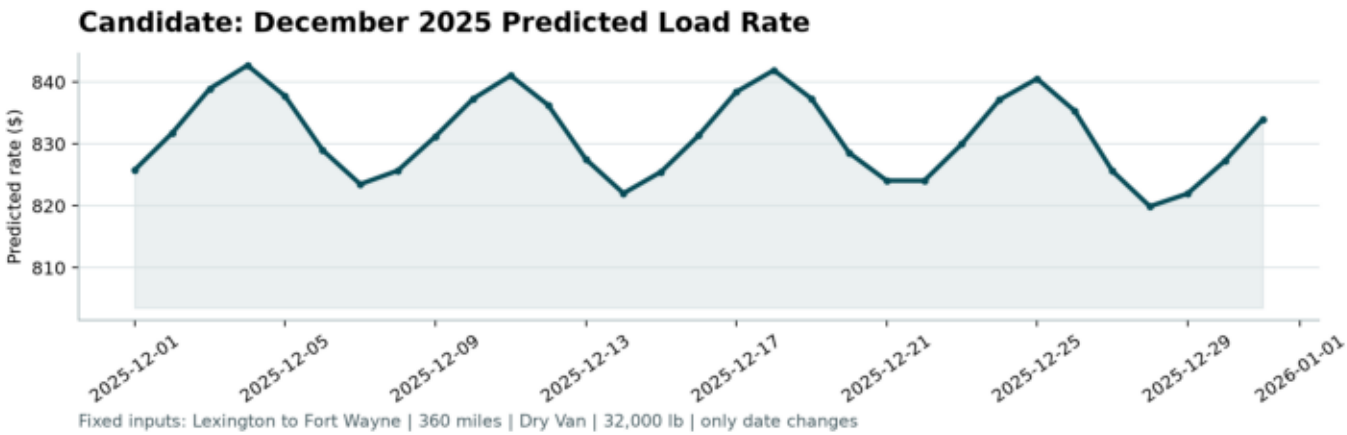
The final models were then refit on every labeled row. Validation predictions ranged from $190.96 to $7,070.74, with median $2,029.51.

# 7. December inference policy

The chart input omits coordinates, market index, and quote signal. Filling those fields from the hidden validation batch would make the chart transductive and would not represent a row-level deployment path. A separate reduced-feature CatBoost model was therefore trained using only pickup, delivery, distance, equipment, repaired weight, and date features. It uses the same robust rate-per-mile formulation.

The source December CSV remained unchanged. Predictions were written to outputs/december_predictions.csv, validated by the supplied scorer, and plotted below.

# 8. Fixed December prediction chart




The fixed-route predictions range from $819.91 to $842.68. Only date changes across these 31 rows.

# 9. Limitations

The final evaluation metric is not disclosed, so the selection score balances MAE and RMSE relative to a common baseline. Extreme target multipliers appear unpredictable from the supplied features and continue to dominate RMSE. The final period also contains unseen cities and a lower market-index distribution. Forward and cold-start validation measure these risks but cannot remove them.

The December model intentionally excludes market conditions because they are absent from the chart schema. Its predictions should not be interpreted as market-aware daily quotes. In a production system, feature availability and timing would be formalized before retraining.